

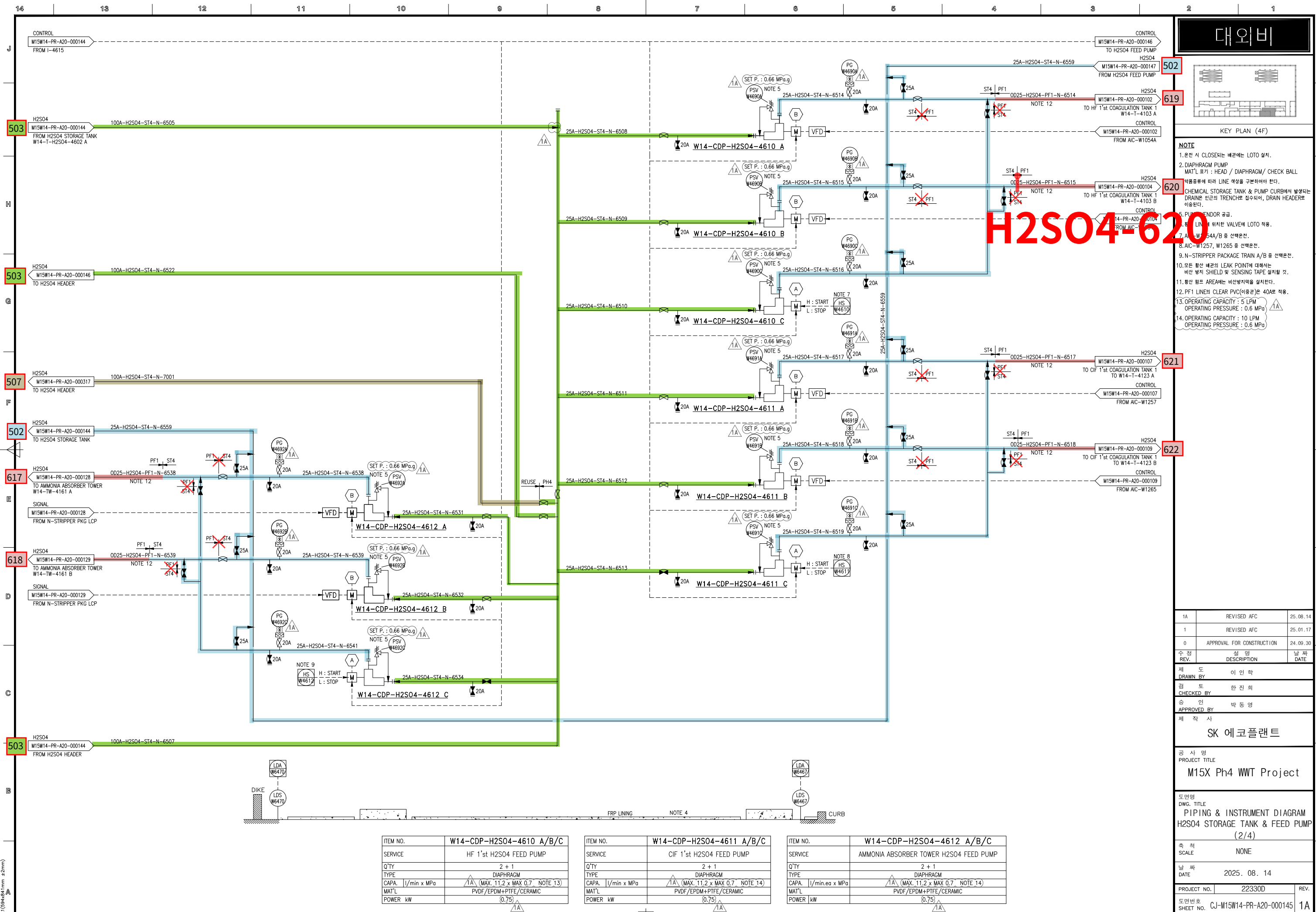
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

H2SO4-620

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	620	4FA	OD25-H2SO4-PF1-N-6515-01	1 OF 2	0	CJ-M15W14-PR-A20-000104	1A	
2	620	4FA	OD25-H2SO4-PF1-N-6515-02	2 OF 2	0	CJ-M15W14-PR-A20-000104	1A	
3	620	5FA	OD25-H2SO4-PF1-N-6515-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
4	620	RFA	OD25-H2SO4-PF1-E-6515-01	1 OF 1	1	CJ-M15W14-PR-A20-000104	1A	



대외비

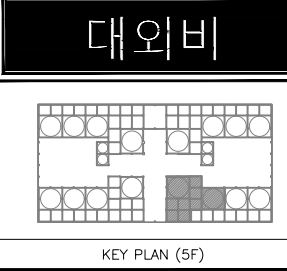
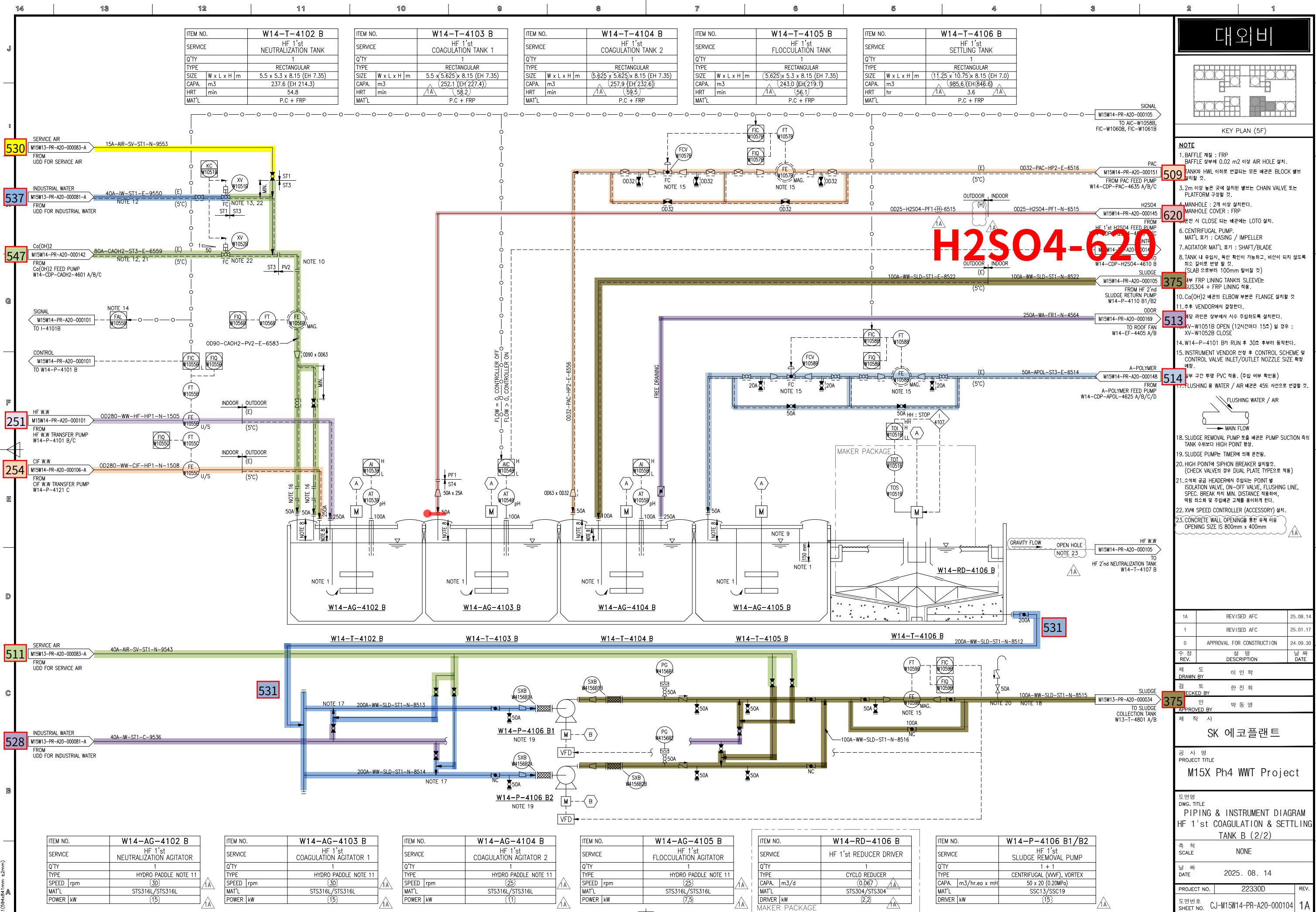
KEY PLAN (4F)

NOTE

1. 온전 시 CLOSE되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 역류종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.
5. PUMP VENDOR 규격.
6. 3.호 LINE에 위치한 VALVE에 LOTO 적용.
7. A/W-WS4A/B 중 선택은전.
8. AIC-W1257, W1265 중 선택은전.
9. N-STRIPPER PACKAGE TRAIN A/B 중 선택은전.
10. 모든 환산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
11. 환산 펌프 AREA에는 비산방지막을 설치한다.
12. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.
13. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa
14. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한진희	
승 APPROVED BY	박동영	
제 제 SK 에코플랜트		
공 PROJECT TITLE	M15X Ph4 WWT Project	
도 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (2/4)	
속 SCALE	NONE	
날 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145	1A

ITEM NO.	W14-CDP-H2SO4-4610 A/B/C	ITEM NO.	W14-CDP-H2SO4-4611 A/B/C	ITEM NO.	W14-CDP-H2SO4-4612 A/B/C
SERVICE	HF 1'st H2SO4 FEED PUMP	SERVICE	CIF 1'st H2SO4 FEED PUMP	SERVICE	AMMONIA ABSORBER TOWER H2SO4 FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 13)	CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)	CAPA. l/min.x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75)	POWER kW	(0.75)	POWER kW	(0.75)



- NOTE**
1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 모든 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. Co(OH)2 배관의 ELBOW 부분의 FLANGE 설치할 것
 11. 주후 VENDOR에서 결정한다.
 12. 해당 라인은 상부에서 시수 투입하도록 설치한다.
 13. XV-W1051B OPEN (12시간마다 15초) 일 경우 :
XV-W1052B CLOSE
 14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 16. 일부 구간 투명 PVC 적용. (주입 여부 확인용)
 17. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 18. SLUDGE REMOVAL PUMP를 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
 20. HIGH POINT에 SIPHON BREAKER 설치할 것.
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
 21. 소액의 공금 HEADER에 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역침 최소화 및 주입배관 고체를 용이하게 한다.
 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 23. CONCRETE WALL OPENING을 통한 유체 이용 OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 회
CHECKED BY	
인	박 동 영
APPROVED BY	
제 작 사	

SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
HF 1st COAGULATION & SETTLING TANK B (2/2)		
속 측	SCALE	
	NONE	
날 짜	DATE	
	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000104	1A



ST4			
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD



1

CONTRACTOR 

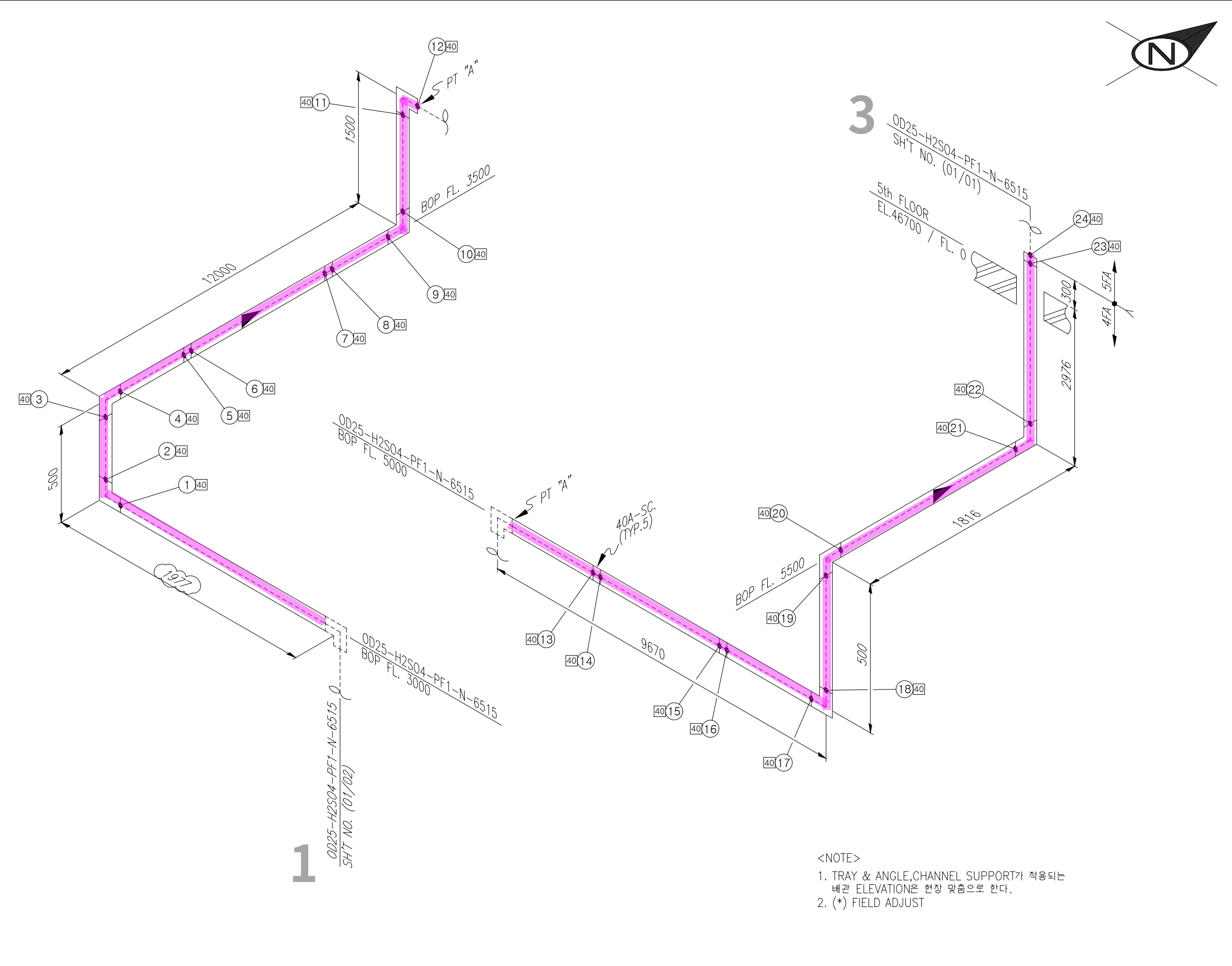
PROJECT NAME
M15X Ph4 Project

COMMODITY	H2SO4
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JOB NO.
22330D

LINE NO.	0D25-H2S04-PF1-N-6515
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SH'T NO.	REV.
01 / 02	0



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	6.9	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A								COMMODITY	H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000145			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD25-H2SO4-PF1-N-6515	SH'T NO. 02 / 02	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

[illegible]

<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

CLIENT



3

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.7	60	UNIT	5FA								M15X Ph4 Project		
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A								COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000144			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D25-H2S04-PF1-N-6515	SH'T NO. 01 / 01	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

